https://edtlab.fr/en/

Engineering Digital Twins (EDT)

Ou *“La Fabrique des Jumeaux Numériques”*

Agence de Programme Numérique
algorithmes, logiciels, usages

RÉPUBLIQUE FRANÇAISE
Liberté Égalité Fraternité

Inria

FRANCE 2030
RESEARCH PROGRAM
Engineering Digital Twins

Program EDT: Co-PIs

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Research Director Inria
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EiC @ SoSyM (Springer Nature)
CSA @ TwiinIT and Myrmix
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edt.community and EDTconf

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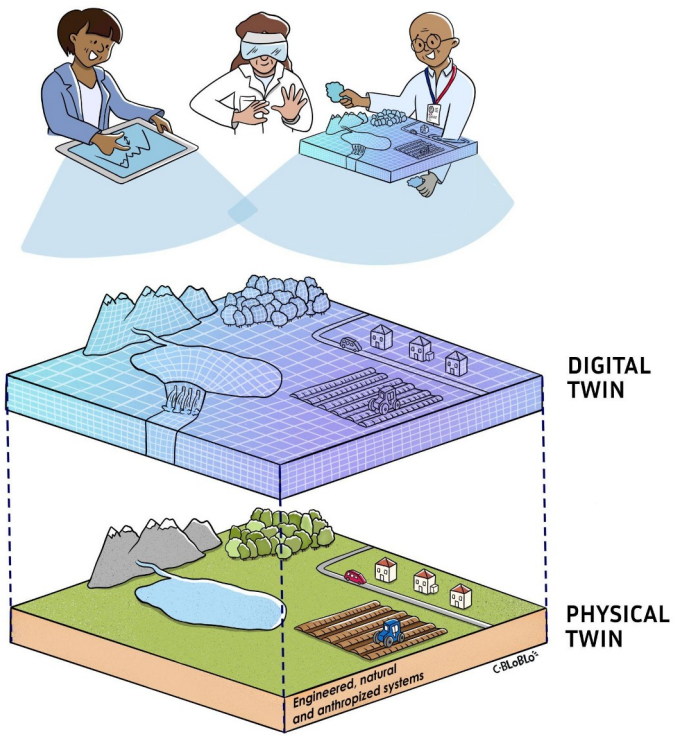
Founder and CEO of Lyatiss and Stackeo
Inria representative at AIF
CS member at RéCLasSIF.

Digital Twins, Yet Another Definition 😅

"A digital twin is a ***software system*** *that* ***purposefully represents*** *an original,* ***accurately reflects*** *its changes, can* ***act upon*** *its original, and provides* ***added value to users or other systems****."*

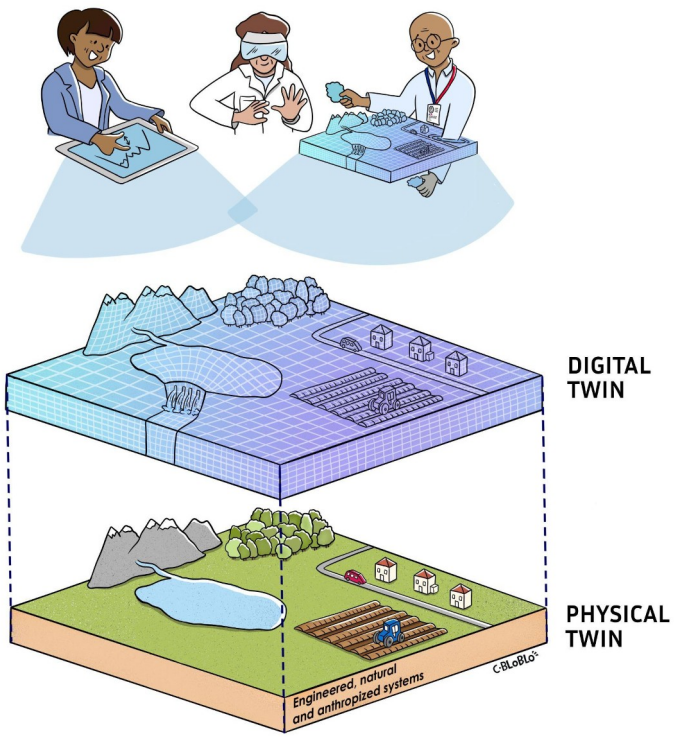
-- Michael, Rumpe, Wortmann, Combemale, Jézéquel.



Digital Twins, My Definition 😅😅

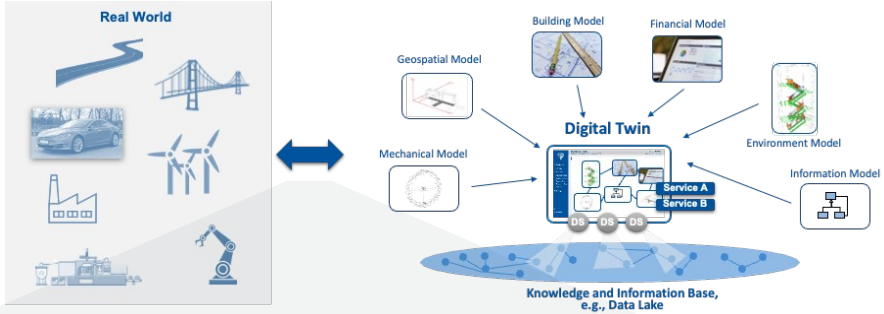
*"A digital twin is a **software system** that **gathers and organizes** information about an original system (, **accurately reflects** its changes, can **act upon** its original,) and provides **controlled access to users or other systems** to use and enrich its information."*

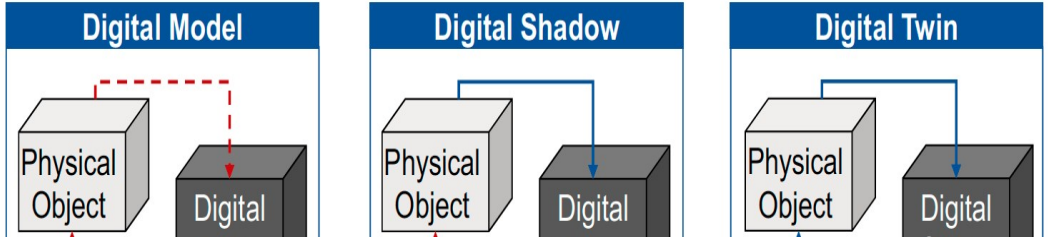
-- Antoine Beugnard.



Creating Digital Twins **of**…

… elements in the "real" world
that can be sensed, monitored,
actuated and controlled



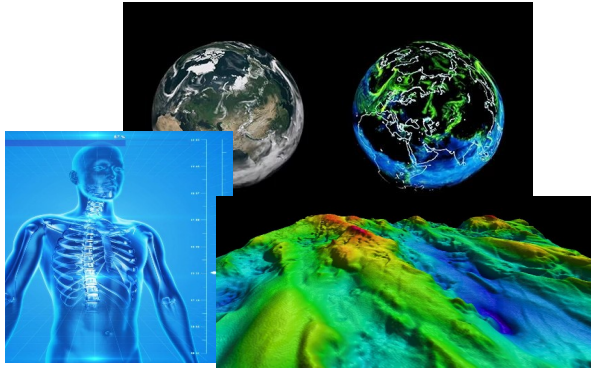


W. Kritzinger, M. Karner, G. Traar, J. Henjes, W. Sihn: *Digital Twin in manufacturing: A categorical literature review and classification.* IFAC-PapersOnLine, 2018.

For…

- observing
- learning
- controlling
- …

Creating Digital Twins **of**…



… natural systems

For…

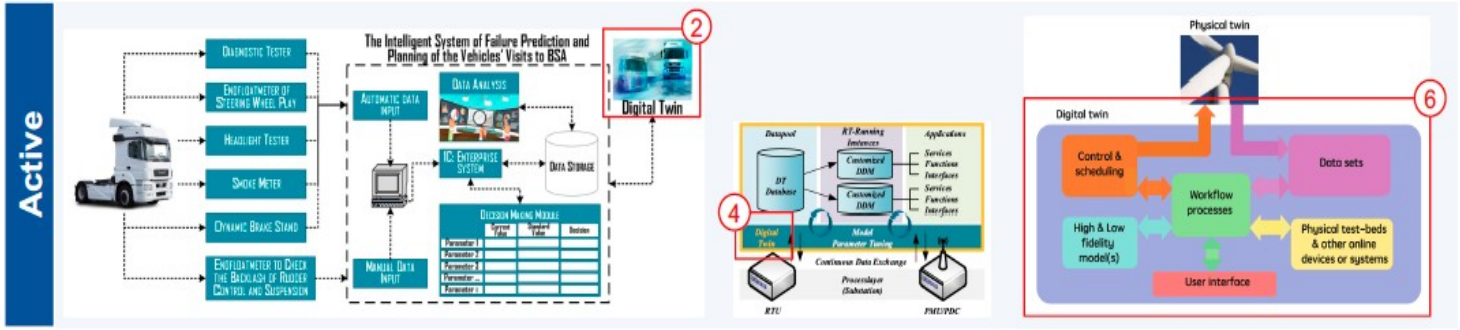
- deciding
- understanding
- …

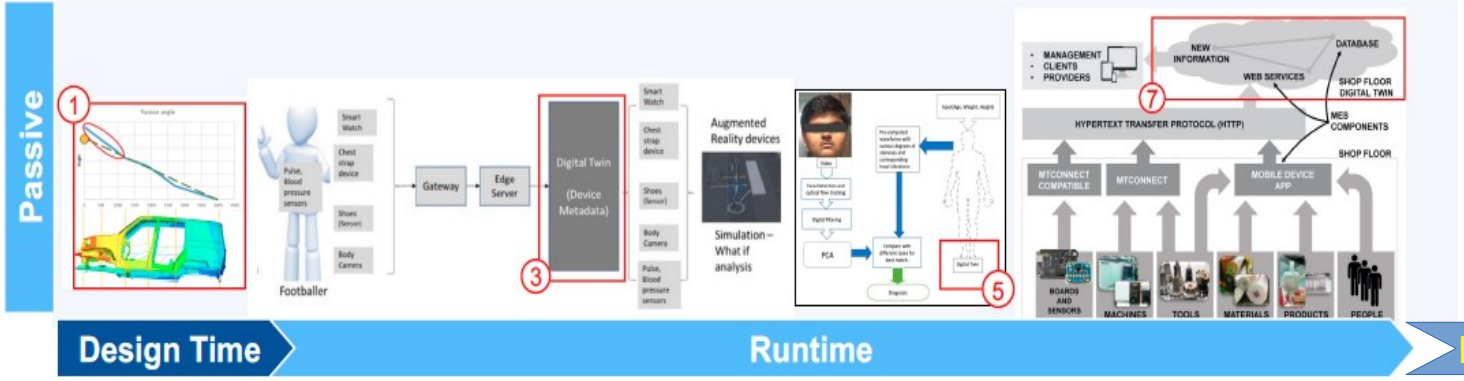
… anthropized systems

… engineered systems

(Cyber-Physical Systems, infrastructures, and business processes)

Creating Digital Twins for…





DT replaces RS

Design Time → Runtime → Decommissioning

Dalibor, Jansen, Rumpe, Schmalzing, Wortmann: *A Cross-Domain Systematic Mapping Study on Software Engineering for Digital Twins*. In: Journal of Systems and Software, 2022.

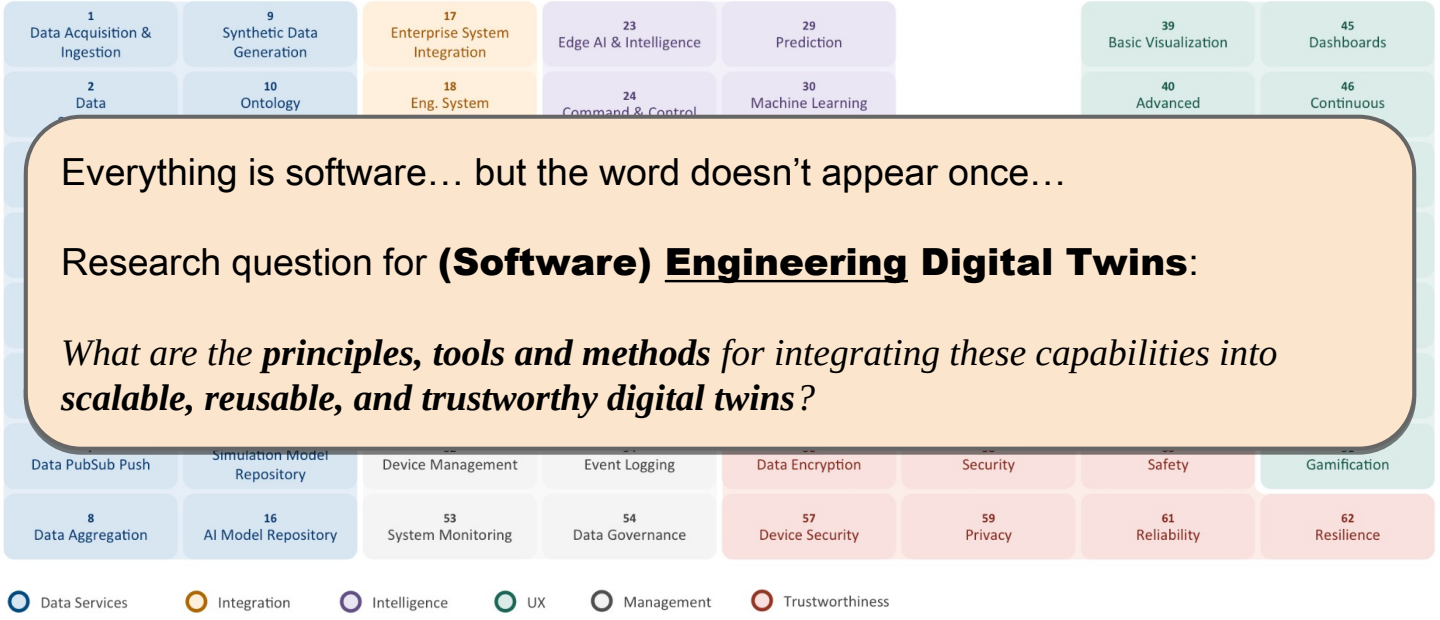
Digital Twin Consortium: Capabilities Periodic Table

1 Data Acquisition & Ingestion	9 Synthetic Data Generation	17 Enterprise System Integration	23 Edge AI & Intelligence	29 Prediction		39 Basic Visualization	45 Dashboards
2 Data Streaming	10 Ontology Management	18 Eng. System Integration	24 Command & Control	30 Machine Learning ML		40 Advanced Visualization	46 Continuous Intelligence
3 Data Transformation	11 Digital Twin (DT) Model Repository	19 OT/IoT System Integration	25 Orchestration	31 Artificial Intelligence AI	35 Prescriptive Recommendations	41 Real-time Monitoring	47 Business Intelligence
4 Data Contextualization	12 DT Instance Repository	20 Digital Twin Integration	26 Alerts & Notifications	32 Federated Learning	36 Business Rules	42 Entity Relationship Visualization	48 BPM & Workflow
5 Batch Processing	13 Temporal Data Store	21 Collab Platform Integration	27 Reporting	33 Simulation	37 Distributed Ledger & Smart Contracts	43 Augmented Reality AR	49 Gaming Engine Visualization
6 Real-time Processing	14 Data Storage & Archive Services	22 API Services	28 Data Analysis & Analytics	34 Mathematical Analytics	38 Composition	44 Virtual Reality VR	50 3D Rendering
7 Data PubSub Push	15 Simulation Model Repository	52 Device Management	54 Event Logging	56 Data Encryption	58 Security	60 Safety	51 Gamification
8 Data Aggregation	16 AI Model Repository	53 System Monitoring	54 Data Governance	57 Device Security	59 Privacy	61 Reliability	62 Resilience

Data Services · Integration · Intelligence · UX · Management · Trustworthiness

https://www.digitaltwinconsortium.org/initiatives/capabilities-periodic-table/

Digital Twin Consortium: Capabilities Periodic Table



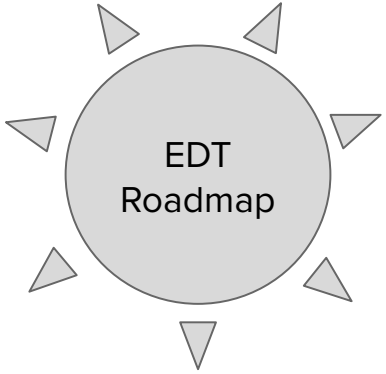
Everything is software… but the word doesn't appear once…

Research question for **(Software) <u>Engineering</u> Digital Twins**:

*What are the **principles, tools and methods** for integrating these capabilities into **scalable, reusable, and trustworthy digital twins**?*

https://www.digitaltwinconsortium.org/initiatives/capabilities-periodic-table/

EDT: Towards a Comprehensive Roadmap



Engineering Digital Twins: A Research Roadmap.
Benoît Combemale, Pascale Vicat-Blanc, Arnaud Blouin, Hind Bril El Haouzi, Jean-Michel Bruel, Julien Deantoni, Thierry Duval, Sébastien Gérard, Jean-Marc Jézéquel. EDTConf, 2025. preprint: https://hal.science/hal-05223776

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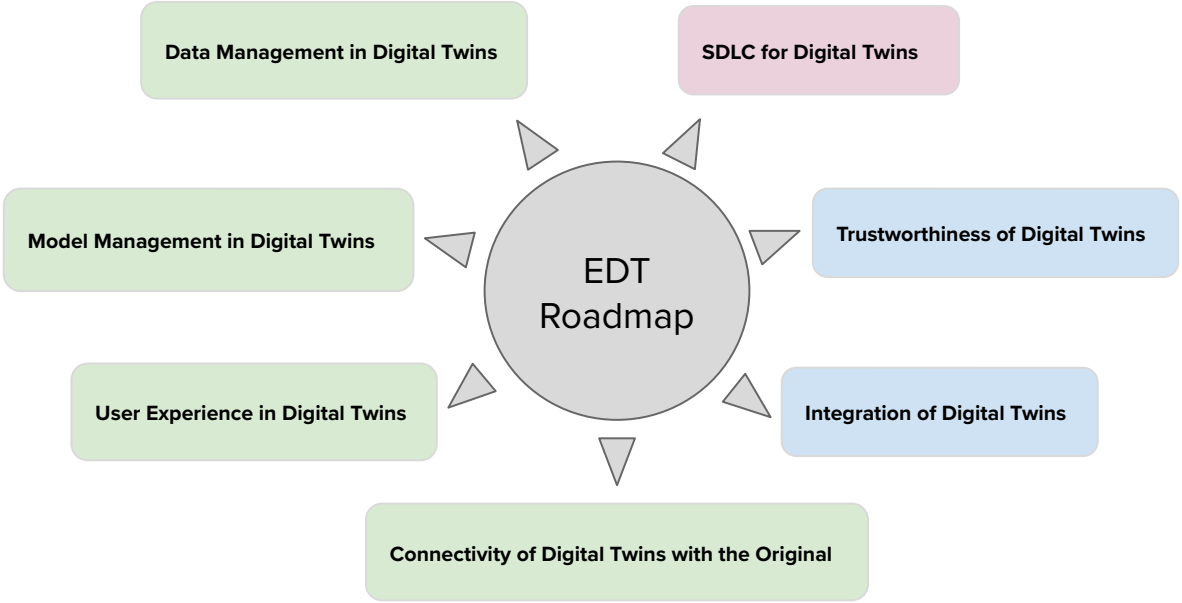

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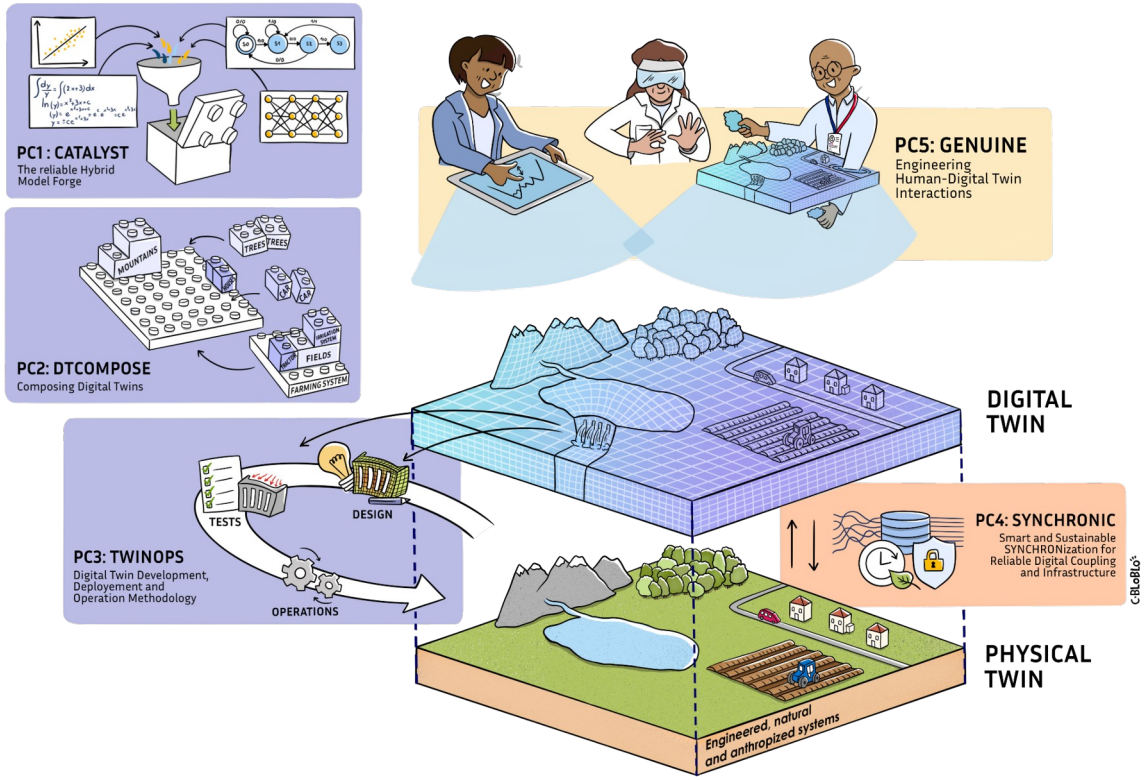
Program EDT: Objectives



EDT National Community

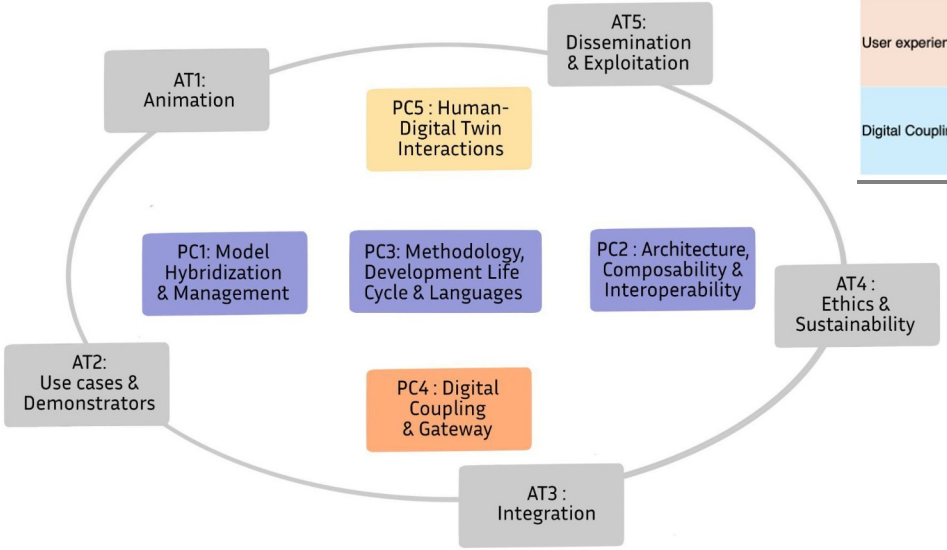
EDT Science Foundations

Artemis DT DevOps platform

Use cases Demonstrators

A nationwide research initiative, joining forces on Software Engineering, Artificial Intelligence, numerical analysis, modeling and simulation, programming languages, Human-Computer Interactions, eXtended Reality, etc.

Program EDT: Implementation



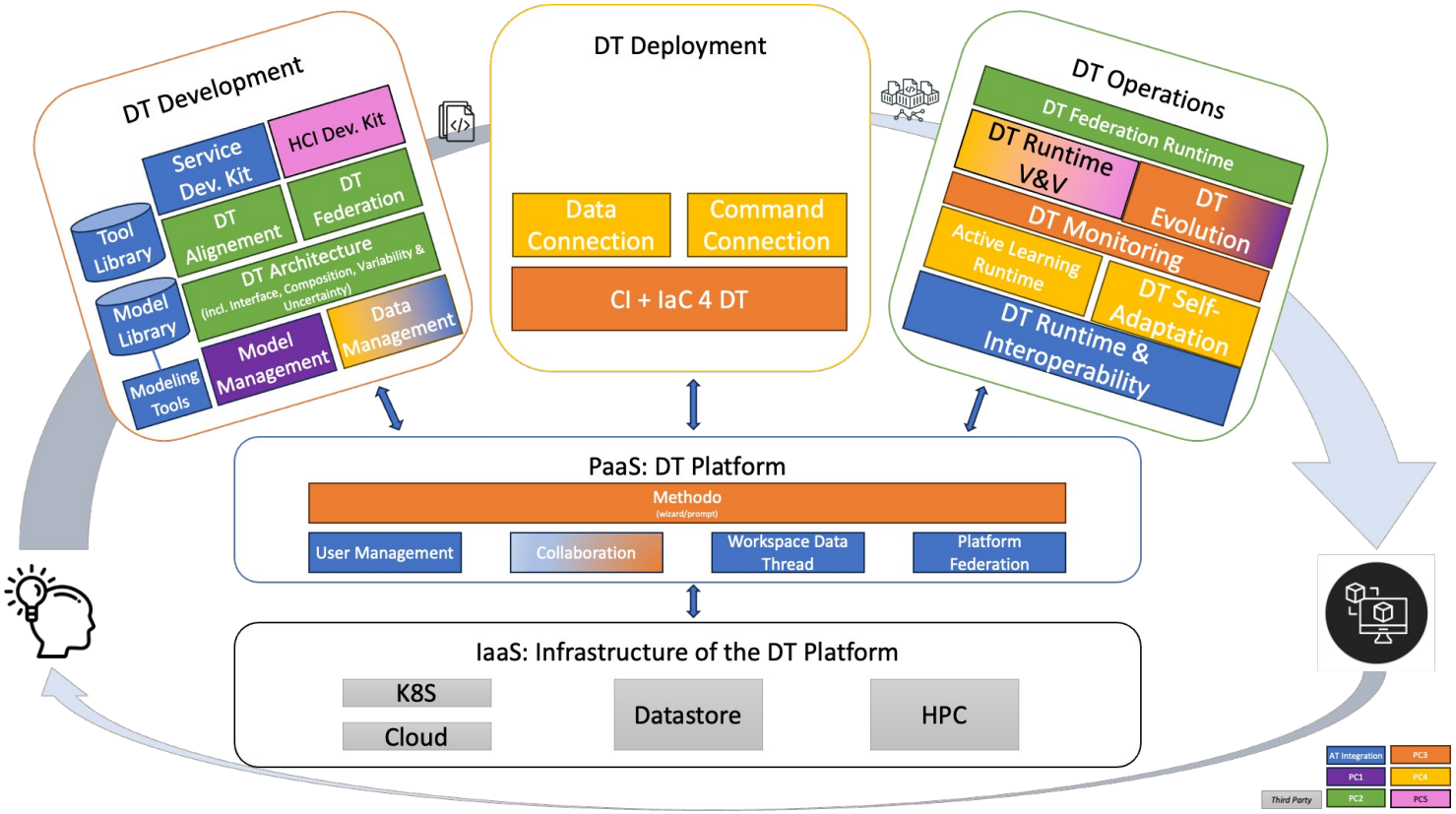
DT Engineering	#	Topics	Keywords
Core Foundations	PC1	Model Hybridization and Management	Model interface for hybrid modeling Abstractions and operators for hybrid modeling Analysis, Validation & Verification
	PC2	Architecture, Composability & Interoperability	DT Reference Architecture and variability Management DT Modularisation, composability and integration DT Interoperability / semantic alignement
	PC3	Methodology, Development Life Cycle & Languages	DT development life cycle and methodology Continuous Engineering / DT evolution Collaborative and augmented development DT Deployment, Frugality and Validation
User experience	PC5	Human-Digital Twin Interactions	Generation of 2D/3D representations from data DT immersive exploitation, interactions and visualization Augmented metaphors (2D/3D, acoustic, haptic…) HCI engineering and co-design Exploration representation
Digital Coupling	PC4	Digital Coupling and Gateway	Smart data collection and assimilation Coupling adaptability Coupling reliability, security and frugality DT-based physical system control Coupling Validation & Verification

Program EDT: The Artemis Platform

- **Low-code DT development**, deployment, maintenance and evolution
  - Data thread management
  - Augmented engineering
  - Continuous engineering
- **Library of reusable models**
- **Connectors to existing tools** (model reduction, simulation, 2D/3D representations, etc.)

⇒ Web-based and collaborative platform to develop and operate digital twins in industry, science and public institutions

Program EDT: The Artemis Platform



Program EDT: Use Cases

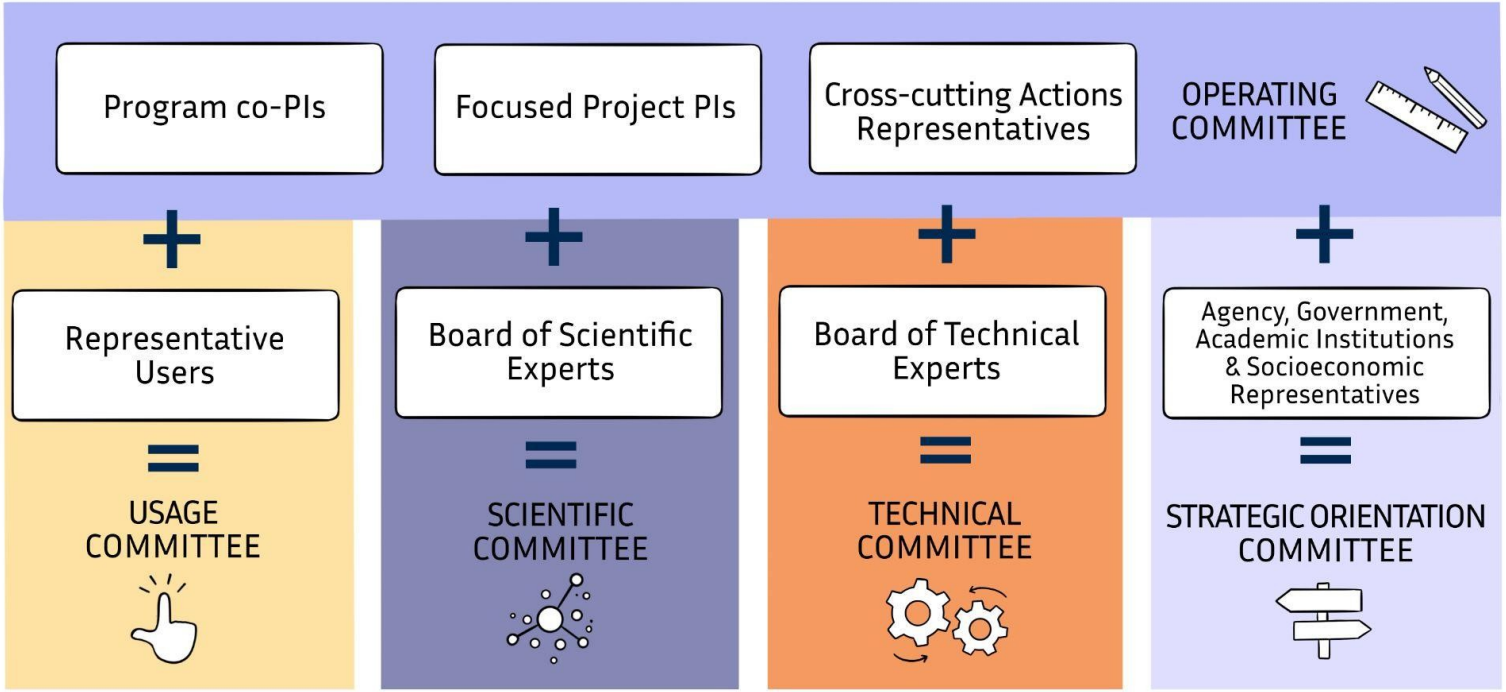
~40 **use cases** provided to the focused projects

~5 selected use cases to provide end-to-end **demonstrators**

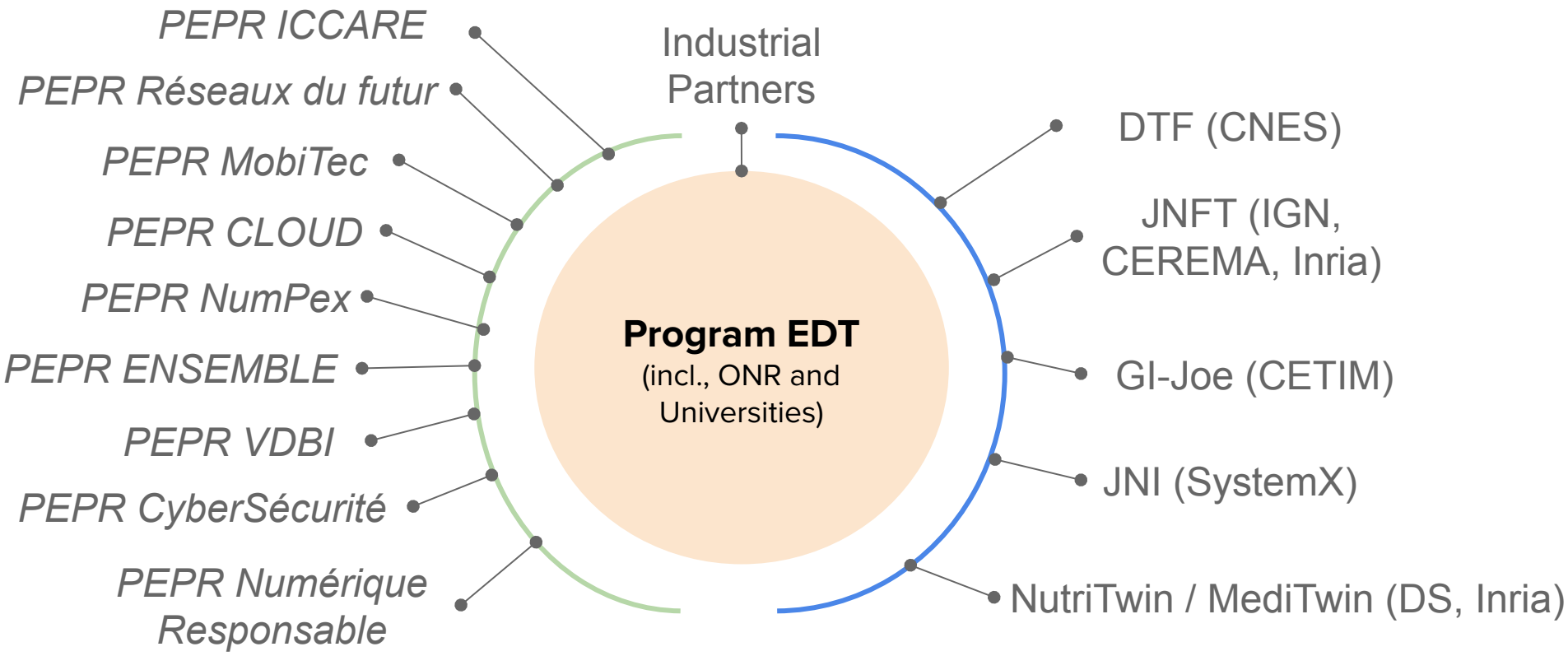
- Natural system
- Anthropized system
- Engineered system (Cyber-physical system, Infrastructure, Business processes)

#	Partners	PT type	UC Topic	UC Title	PC1	PC2	PC3	PC4	PC5
UC1	INRAE	Natural	Biology	Microbiot	X			X	X
UC2	IFPEN	Natural	Environment	Pollution detection	X			X	
UC3	CNES	Natural Anthropized	Climate	Climate Change Adaptation	X	X	X	X	X
UC4	ONIRIS	Natural Anthropized	Climate	GreenHousesControl	X			X	X
UC5	INRAE	Natural Anthropized	Agro-ecology	InSiliCow - Smart Farms	X	X	X	X	X
UC6	URCA CIRAD INRAE	Natural Anthropized	Agro-ecology	Virtual Plant - OmiPl@nt	X				X
UC7	CIRAD	Natural Anthropized	Agro-systèmes	Production systems for agro-ecologic transition	X	X	X	X	X
UC8	INRAE	Anthropized	Mobility	Mobility on a territory	X	X		X	
UC9	IGN CNRS	Anthropized	Territory	Complex territory management	X	X	X	X	X
UC10	IRD	Anthropized	Territory	Impact of urban interventions in developing countries					X
UC11	INRIA	Anthropized	Territory	Temporary layout design of public places	X				X
UC12	INRAE	CPS	Robotics	Robotics for Agro-ecology (safety, power, localisation...)	X	X	X	X	X
UC13	IFPEN	CPS	Energy	Offshore Wind Turbine Farm	X	X		X	
UC14	CEA	CPS	Industry	Industrial Metaverse				X	X
UC15	CEA	CPS	Industry	DT for training			X		X
UC16	CEA	CPS	Industry	Photorealistic DT	X				X
UC17	CEA	CPS	Industry	Prism - Industry 5.0 testbed			X	X	X
UC18	IMT	CPS	Industry	Industrial Maintenance	X	X	X	X	X
UC19	CNRS GSCOP	CPS	Industry	Co-engineering of an aerospace product-process system of systems	X	X	X		X
UC20	CNRS PSL	CPS	Industry	Reliability and sustainability of composite materials	X			X	
UC21	IMT	CPS	Industry	Bioreactors	X	X	X	X	
UC22	IMT INRIA	CPS	Industry	ManufacturingFactory	X	X	X	X	X
UC23	U.Eiffel INRIA	CPS	Transport	CityVal Line B	X			X	X
UC24	INRIA CNRS	CPS	Transport	Drones for smoke or fire detection missions			X	X	
UC25	UPHF CNRS	CPS	Transport	DT of Fleet of trains		X	X		
UC26	UPHF CNRS	Infrastructure	Transport	RailDIT	X		X		X
UC27	IFPEN	Process	Industry	Low Carbon Industrial Process	X				
UC28	CEA	Process	Industry	Multimodal terminal	X				X
UC29	CEA	Process	Health Care	Hospital Department	X				
UC30	CEA	Process	Logistic	Warehouse	X				X
UC31	UB	Process	Culture	Anastylosis	X	X	X		X
UC32	UB CNRS	Process	Education	Personalised Learning Pathway	X			X	

Program EDT: Implementation

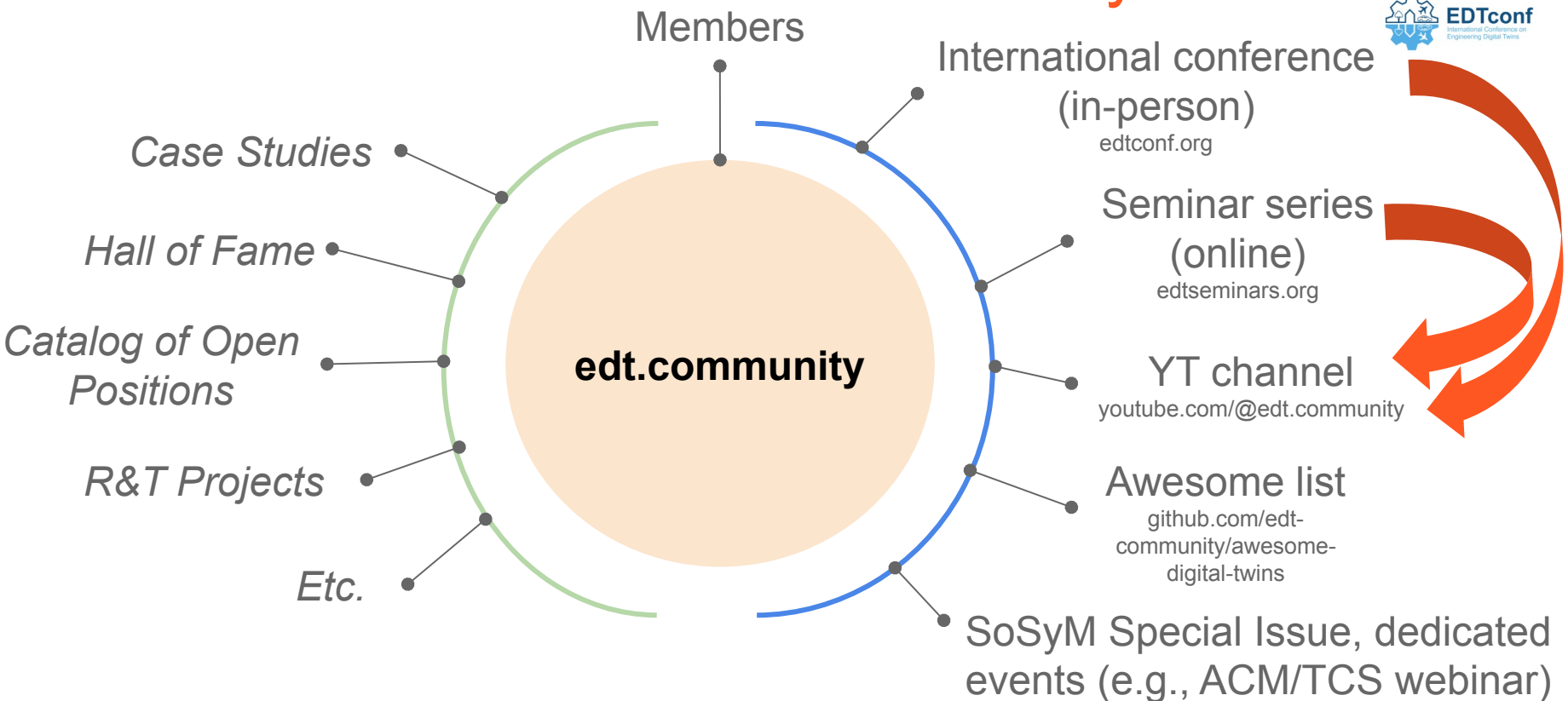


Program EDT: Ecosystem



https://edt.community

Engineering Digital Twins (EDT): An International Scientific Community



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